Shaheed Zulfikar Ali Bhutto Institute of Science & Technology University

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 7.5

Obtained Marks:__________

COAL (Lab)

Final Report

ATM Transaction Analysis

Last Date of Submission: 21-05-2026

Submitted To: **Muhammad Hasaan Mujtaba**

Student Names	Registration No
Azaam Zameer	24108467
Abdul Rehman	24108345

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

ATM Transaction Simulator

Introduction :

Automated Teller Machines (ATMs) play a vital role in modern banking by enabling customers to perform essential financial transactions such as cash withdrawal, deposits, and balance inquiries without human assistance. Although these systems appear simple to users, internally they rely on highly secure, accurate, and time-critical software. Understanding how such systems work at a low level is important for students of Computer Organization and Assembly Language, as it reveals the direct interaction between hardware and software.

The **ATM Transaction Simulator** project is developed using **8086 Assembly Language** to provide a practical understanding of real-world banking system operations. This project focuses on low-level concepts such as register usage, segmented memory architecture, interrupt-driven input/output, flag-based decision making, and secure data handling. By implementing ATM functionalities in assembly language, students gain hands-on experience with how financial systems can be built under strict hardware and memory constraints.

Abstract :

This project presents the design and implementation of a secure ATM Transaction Simulator using 8086 Assembly Language. The simulator demonstrates key banking operations including PIN authentication, balance inquiry, withdrawal, deposit, and transaction logging. The system emphasizes secure input handling, transaction validation, and memory management using the Intel 8086 architecture. By using processor registers, flag monitoring, and DOS interrupt services, the project bridges theoretical computer organization concepts with practical financial system design. The simulator serves as an effective educational tool for understanding low-level programming and security-aware system development.

Problem Statement & Constraints :

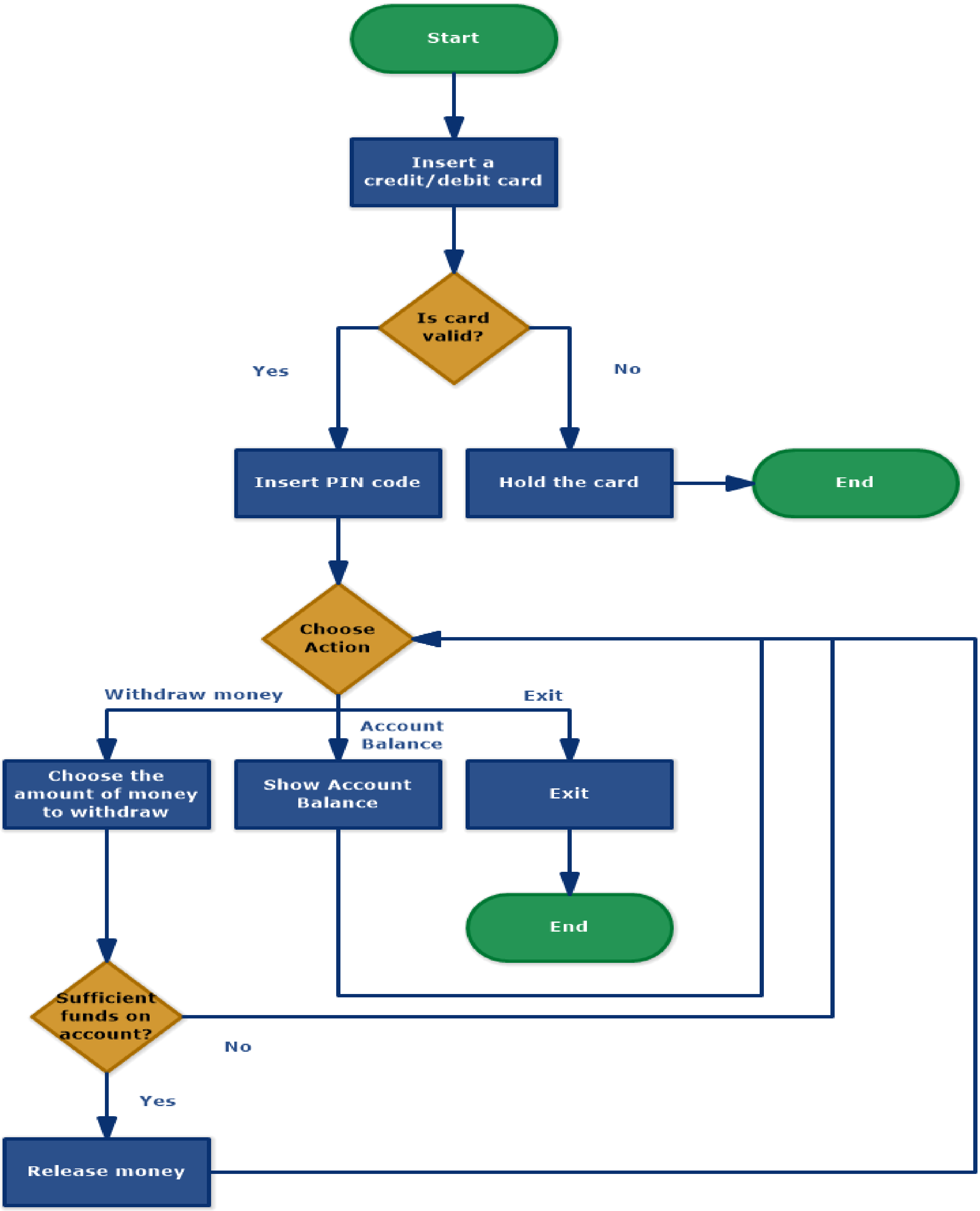
Problem Statement :

How can a secure and functional ATM transaction system be designed using 8086 assembly language that demonstrates authentication mechanisms, transaction integrity, efficient memory usage, and real-time user interaction while remaining suitable for educational purposes?

Constraints :

- Limited memory (64 KB per segment)
- 16-bit registers and arithmetic limitations
- Text-based interface using DOS interrupts
- Single-user and single-session operation
- No persistent storage (data lost on program termination)
- Platform dependency on 8086 architecture and emulators

Flowchart & Algorithm :



Algorithm Steps :

1. Program starts and initializes data and stack segments
2. User is prompted to enter PIN securely
3. Entered PIN is compared with stored PIN
4. If PIN is incorrect, attempts counter is decremented
5. Account is locked after three failed attempts
6. If PIN is correct, main menu is displayed

7. User selects transaction option
8. System validates input and performs selected operation
9. Balance is updated atomically
10. Transaction is logged
11. Control returns to main menu or program exits

Register & Memory Mapping :

Register Usage :

- AX / AL: Arithmetic operations and data transfer
- BX: Base register for array indexing
- CX: Loop counter
- DX: I/O operations and remainder storage
- SI: String and array traversal
- SP / BP: Stack handling

Memory Organization :

- Data Segment: PIN, balance, transaction history
- Code Segment: Program instructions
- Stack Segment: Temporary storage and rollback support

Instruction Set & Addressing Modes Used :

Instruction Set :

- MOV, LEA, PUSH, POP
- ADD, SUB, MUL, DIV
- INC, DEC
- CMP, XOR
- JMP, JE, JNE, LOOP
- INT 21H

Addressing Modes :

- Immediate addressing
- Direct addressing
- Indirect addressing

- Base-displacement addressing
- Indexed addressing

The ATM Transaction Simulator was tested using emu8086 and DOSBox environments. The simulation successfully demonstrated secure PIN masking, correct authentication logic, accurate balance updates, overflow prevention during deposits, and proper transaction logging. The menu-driven interface responded correctly to user input under all tested scenarios.

Flag & Register State Analysis :

- Zero Flag (ZF): Used for PIN matching and validation
- Carry Flag (CF): Used for overflow and underflow detection
- Sign Flag (SF): Used for signed arithmetic
- AX Register: Stores arithmetic results
- CX Register: Controls loop execution

Challenges & Solutions :

Challenges :

- Secure input handling without high-level functions
- Decimal to binary conversion
- Overflow detection
- Limited registers

Solutions :

- Used INT 21H (07H) for secure input
- Implemented digit-by-digit conversion
- Monitored Carry Flag for overflow
- Used stack operations for register preservation

Conclusion :

The ATM Transaction Simulator successfully demonstrates how real-world banking system concepts can be implemented using 8086 assembly language. The project strengthens understanding of computer organization principles, low-level security mechanisms, and transaction processing. It provides valuable hands-on experience and prepares students for advanced studies in embedded systems, operating systems, and cybersecurity.